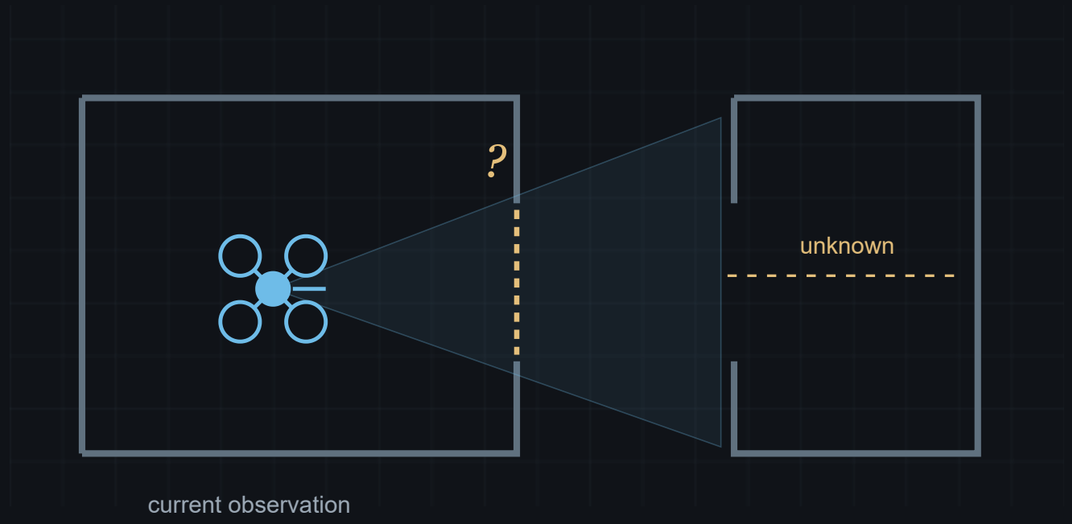
The course is visible.
The next action is uncertain.

Indoor navigation, incomplete observations, and a mission beyond the field of view.



What should happen next?

A passage candidate

What lies beyond it?

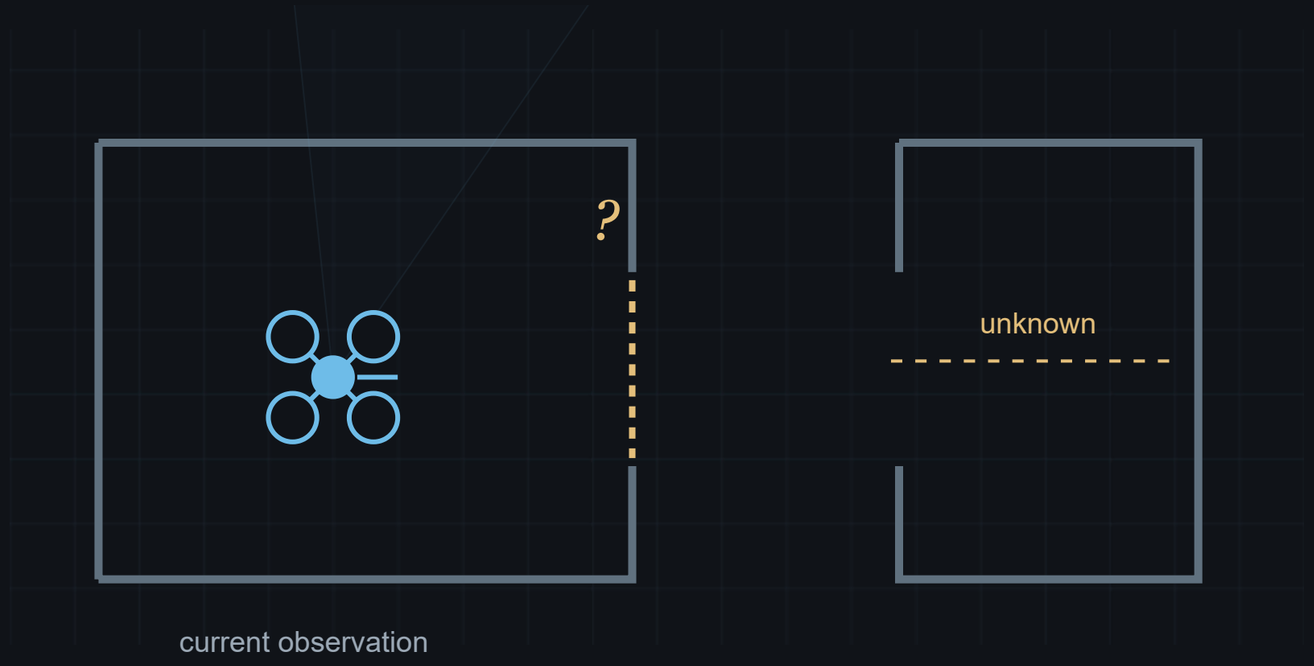
Observe before committing.

geometry + memory + intent

↳ Each camera frame reveals only part of the task.

Build what the drone *remembers*.

Persistent identity. Evidence. Uncertainty. A reason to observe again.



Memory ≠ a fresh view

opening_0003

same entity, new viewpoint

last seen → aging

identity persists · certainty does not

↳ Prioritize a useful world representation before increasing policy complexity.

Illustrative belief update

Combine ideas at the interfaces.

Research gave us the structure; our constraints shaped the implementation.

REPRESENTATION

Hydra

incremental geometry → places

Kimera

metric + semantic structure



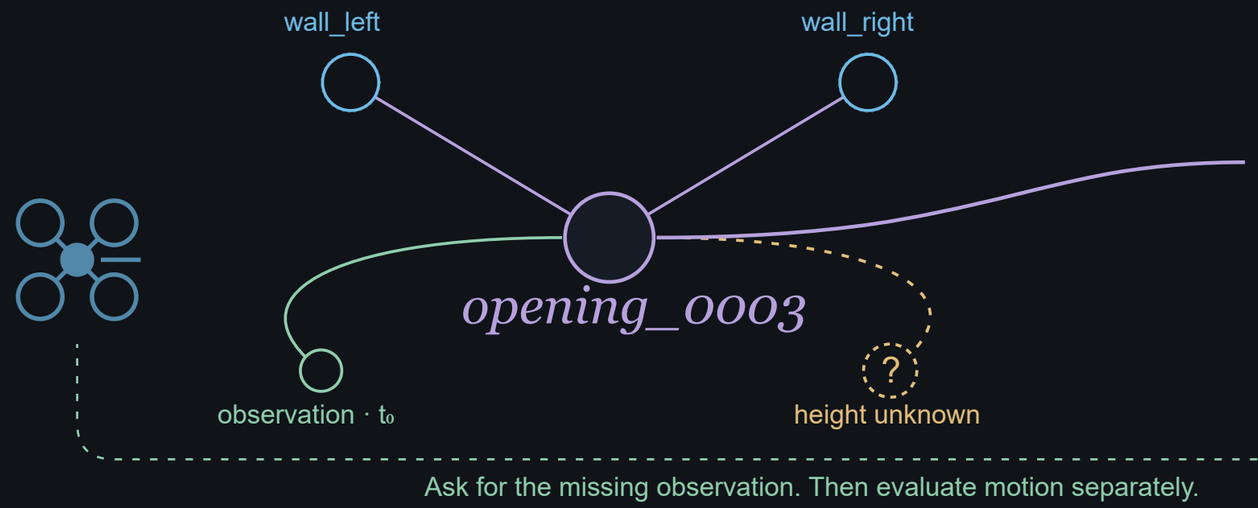
Adapt the ideas. Measure our own system.

↳ A compact adaptation for our task and compute budget.

From seeing to *understanding*.

Observations become geometry, then meaning, then a grounded decision.

01 OBSERVATIONS | 02 GEOMETRY | 03 SEMANTIC GRAPH | 04 REASONING



GOAL PROPOSAL

observe_more

target: opening_0003

because: height is unknown

Grounded proposal ≠ permission to fly

Ask for the missing observation. Then evaluate motion separately.

↳ Missing evidence becomes a reason to look again.

Illustrative entity · offline contract

Sometimes the next action is *to look*.

A bounded reasoning layer turns missing information into an explicit goal.

WHAT WE KNOW

opening_0003

width observed

height unknown

last observation retained

→

Bounded context

relevant entities

evidence + missing facts

→

Goal proposal

observe_more

target: opening_0003

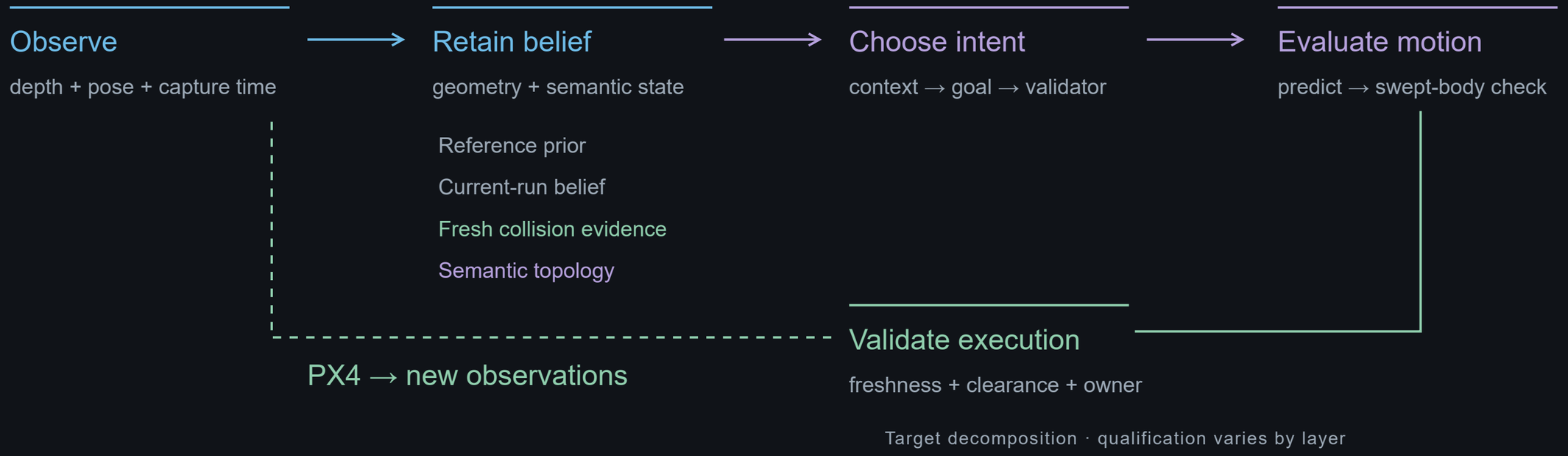
Validate identity, freshness and grounding.

A valid proposal still needs motion evaluation.

↳ “Observe this opening again” is a useful decision.

Each layer answers a different question.

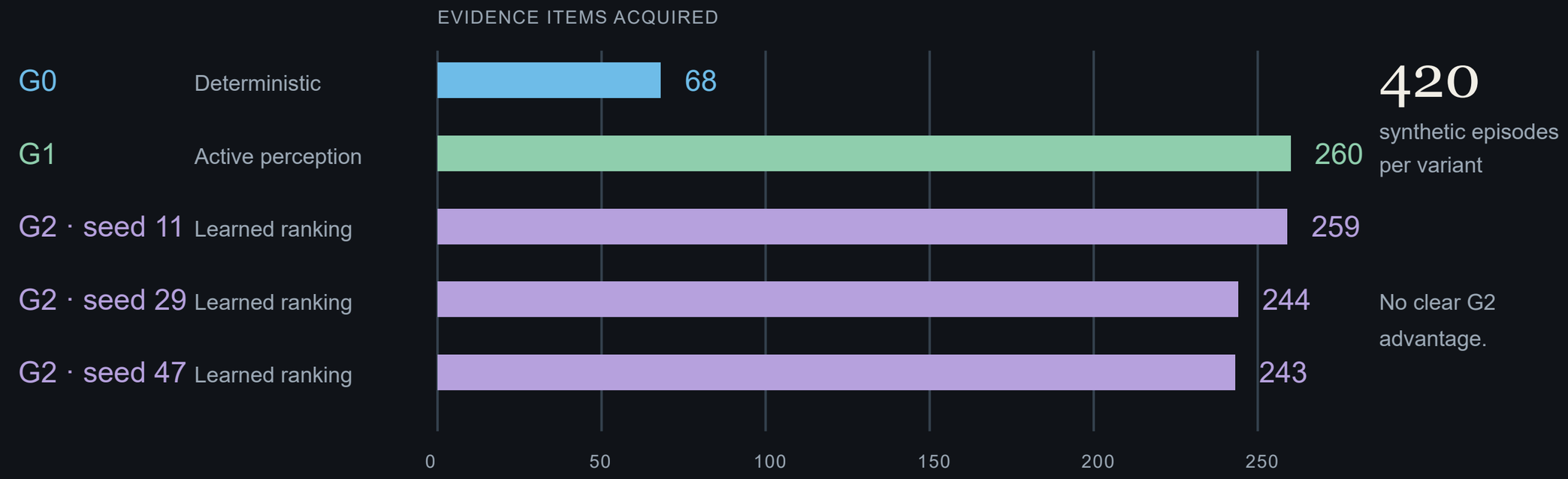
Our target architecture separates memory, intent, motion and execution.



↳ Persistent belief and fresh collision evidence have different jobs.

Complexity has to earn its place.

Active perception earned priority in a synthetic evidence-acquisition task.



420
synthetic episodes per variant

No clear G2 advantage.

↳ Active perception improved this task; learned ranking showed no clear advantage.

Synthetic benchmark · not flight success

A better map was not yet a better decision.

An integration lesson: better estimates must reach the decisions they are meant to change.

EARLY P2 FIXTURE

Better map pose

localization error improved

×

Same stops

planner did not consume it

The missing link was a contract.

Bounded worker → Diagnostic consumer → Decision benefit: unproven

Later offline integration · historical progression

↳ We shifted toward explicit consumer contracts and bounded asynchronous registration.

The assumptions failed between layers.

Native experiments exposed altitude, frame, timing and scheduling contracts.

C1 · NATIVE CALIBRATION

The test never reached braking.

Translate ⟶ Height gate ⟶ LAND

Measurement failed. Landing and disarm completed.

→ Altitude retention

Schematic event sequence · not a telemetry curve

S0 · NATIVE COURSE ATTEMPTS

Rollout ≠ native flight

01 frame origin

02 timestamp skew

03 readiness reset

04 scheduling

Attempts 001–004 stayed disarmed.

→ Frame binding + bounded work

↳ Preserved failures became specific engineering work.

What we built. What we still need to prove.

Evidence is specific to a component, an experiment and an environment.

Offline evidence

- Persistent entities + reasoning
- Synthetic active-perception gain
- Mapping + diagnostic integration

Native evidence

- Disarmed observations
- C0 takeoff / hover / landing
- C1 recovery; measurement failed

Still to demonstrate

- Integrated hybrid course success
- Matched navigation benefit
- Hardware qualification

Separate baseline

Classical S1: one successful Gazebo course run.

↳ The hybrid architecture has not yet demonstrated full-course native success in the fetched record.

From observations to *justified actions*.

The next milestone is a measurable mission benefit from each layer.

01

Reliable native segment

frames · timing · height control

→

02

Matched comparison

same map, start and scoring

→

03

Layer ablations

memory → topology → semantics

Observations → *Grounded belief* → *Justified actions*

Learned extensions only when the evidence warrants them.

↳ Our contribution is a testable path from observations to justified actions.

N1 · N6 · N7 · N8

Planned evaluation

A map can say where. Semantics asks what next.

Three representations

- **Geometry:** surfaces, pose, observed free space and clearance.
- **Topology:** places and supported connections.
- **Semantics:** task-relevant meaning, such as a candidate passage.

Why keep memory?

Retain identity when a feature leaves view. Track stale or contradictory evidence. Select an information-gathering goal rather than inventing a connection.

These are architectural capabilities; mission benefit still needs a matched ablation.

Learning is a replaceable component.

Reasoning comparison

- G0: deterministic baseline.
- G1: utility / information-gain selection.
- G2: compact learned candidate ranking, evaluated with three seeds.

Motion prediction

D0 is the nominal physics default. Residual and adaptive alternatives exist offline, but clean-case improvements did not satisfy every delay, heading and runtime gate.
A learned predictor is not a learned navigation policy or a VLA controlling the aircraft.

Compare the same task, under the same conditions.

Reasoning benchmark

60 held-out layouts × 7 regimes × 6 steps. 420 episodes per variant.
Evidence acquired: G0 68; G1 260; G2 259 / 244 / 243.
Abstract synthetic evidence acquisition, not course completion.

Separate Classical S1 result

One Gazebo run: four checkpoints, pad landing and disarm in 84.3 seconds; zero touches reported.
Run `run-AdMalPxA`, revision `ad31ae5`. Not a matched hybrid comparison or physical-hardware result.

A useful fact needs a frame and a clock.

Data contracts

- Capture time, frame, units and reset epoch.
- Stable identity and supporting observations.
- Bounded context and independent validation.
- One command owner.

Four separate layers

- Permitted reference geometry.
- Persistent current-run belief.
- Fresh local collision evidence.
- Semantic and topological state.

A prior map does not become a fresh sensor observation.

A recovered experiment can still be a failed test.

Preserved outcomes

C1 crossed its height threshold before the brake trigger. LAND and disarm completed, but no braking envelope was measured. Latest fetched S0 attempts 001–004 stayed disarmed. Frame, timestamp and scheduling work remained.

Acceptance sequence

Native segment → matched baseline → memory / topology / semantics ablations → changed-layout evaluation.
Measure mission progress, false-feasible choices, observation cost and latency. Add learned complexity when these improve.

The ideas we adapted.

Academic references

Hydra · Hughes et al. · RSS 2022
Incremental hierarchical scene graphs.

Kimera · Rosinol et al. · 2021
Metric-semantic spatial representation.

CSIRO Data61 · Hudson et al. · 2021
Fielded subterranean robotic systems.

Engineering and visual references

ModalAI VOXL Vision Hub
VIO, obstacle processing and autopilot interfaces.

3Blue1Brown / Manim
Explanatory animation inspiration. All diagrams here are original SVG illustrations.

Methods inform our design; their published results do not validate our platform.